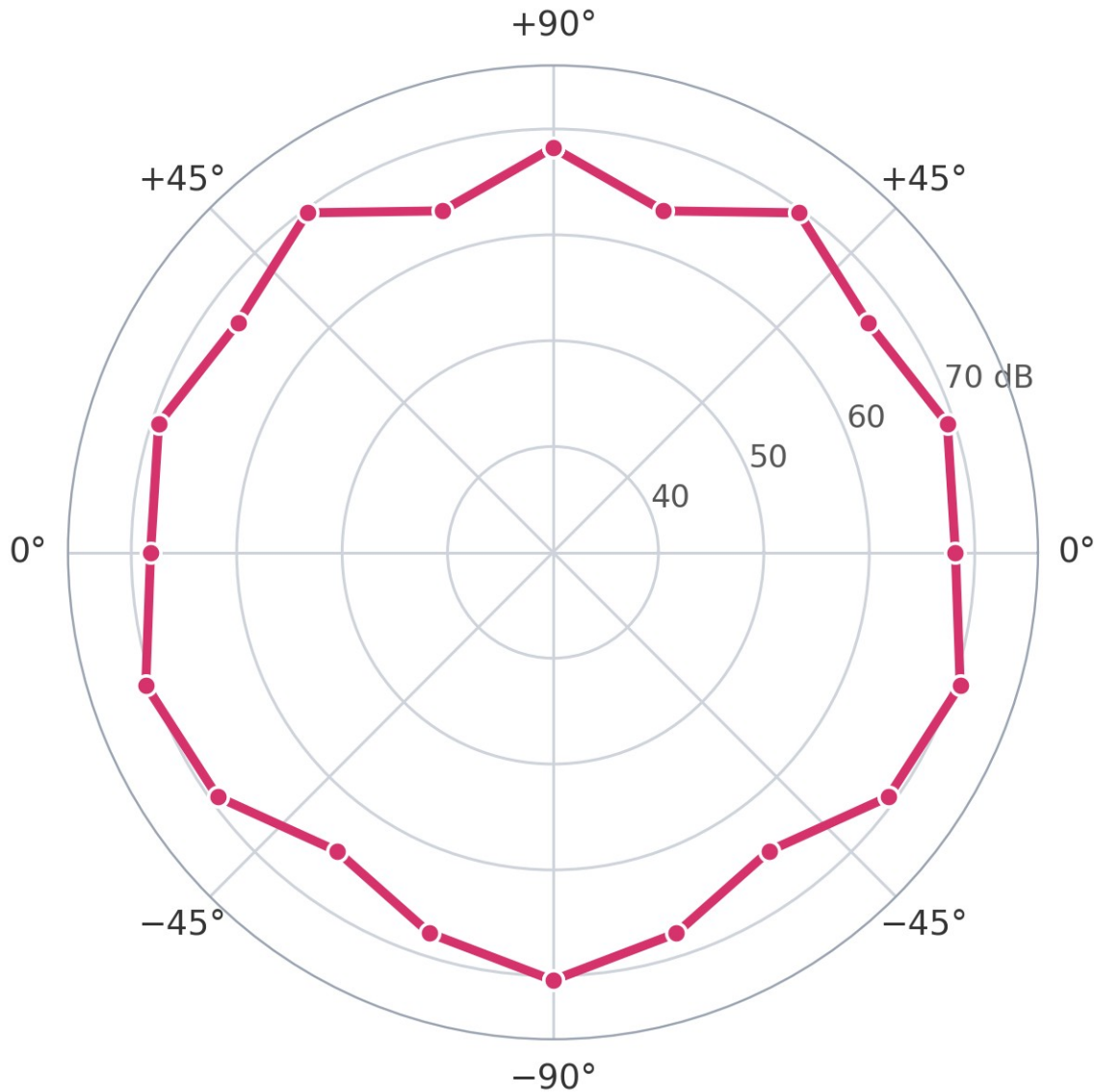




Chasing a round polar

Our chamber in its current set-up (5 Oct 2026): eleven microphones on a ring around the source, absorbing material on floor, ceiling and walls.



THE PROBLEM

Our polar plot is jagged

- We measured a propeller with the microphone arc in a plane parallel to the propeller disc.
- A propeller is symmetric around its axis, so every microphone should read about the same.
- **We expected a smooth, nearly round polar. We observed a jagged outline.**

6.7 dB

between the loudest and quietest of the eleven microphones

THE FIX

What we did: every microphone, one loudspeaker



Our loudspeaker

tones, 62 Hz–16 kHz



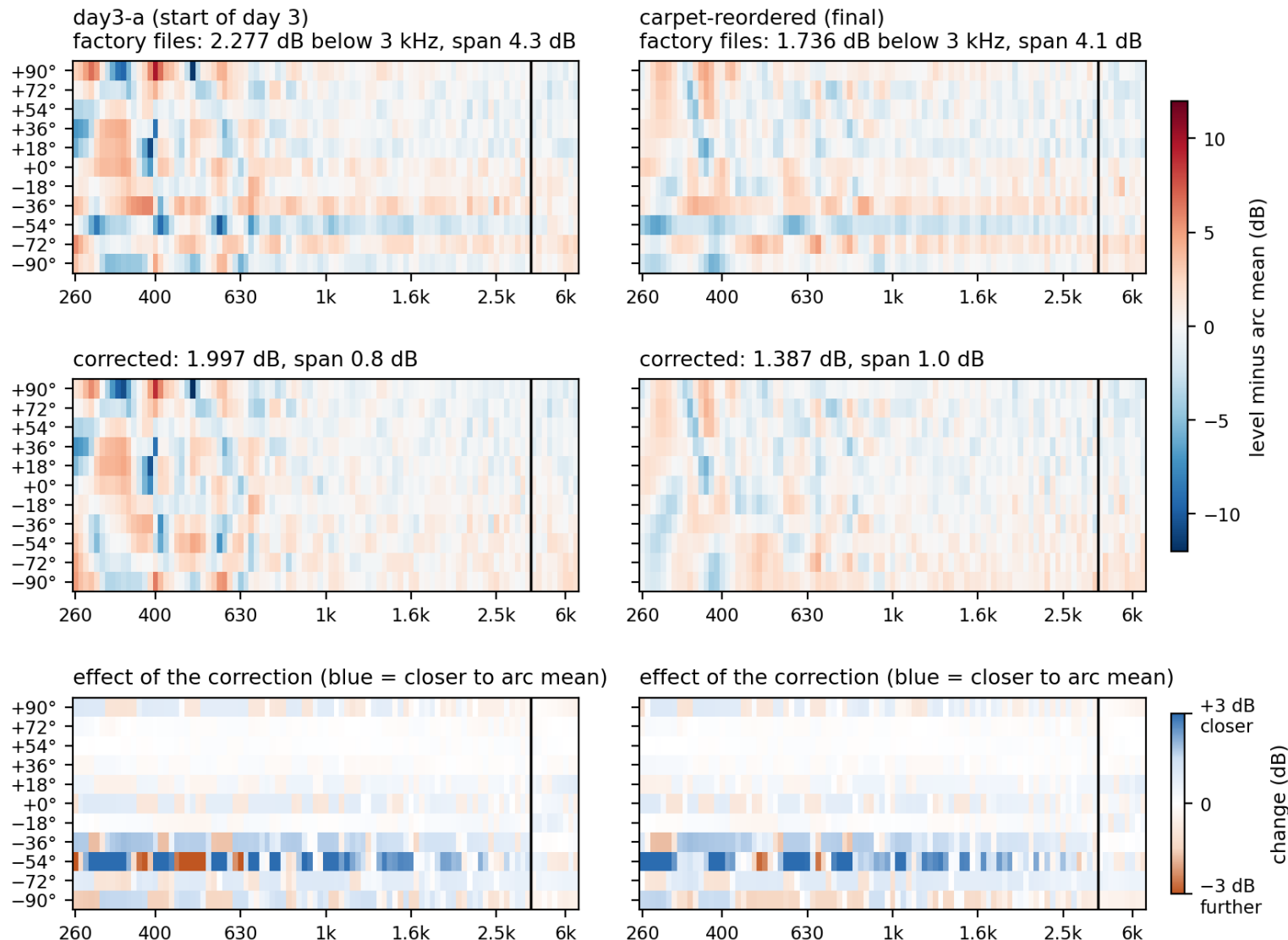
UMIK-2 microphone, capsule faces the loudspeaker

- 97 tones per microphone, 62 Hz to 16 kHz, 12 per octave (a stepped sweep)
- One tone at a time: 0.7 s to settle, then 2 s recorded (6 s below 250 Hz)
- Microphones swapped in turn: same position, same loudspeaker, same level
- One microphone measured three times as the reference

4.01 → 0.02 dB

spread across the eleven microphones for the same sound

7 of 11 factory files were off by 0.8–2.6 dB on average



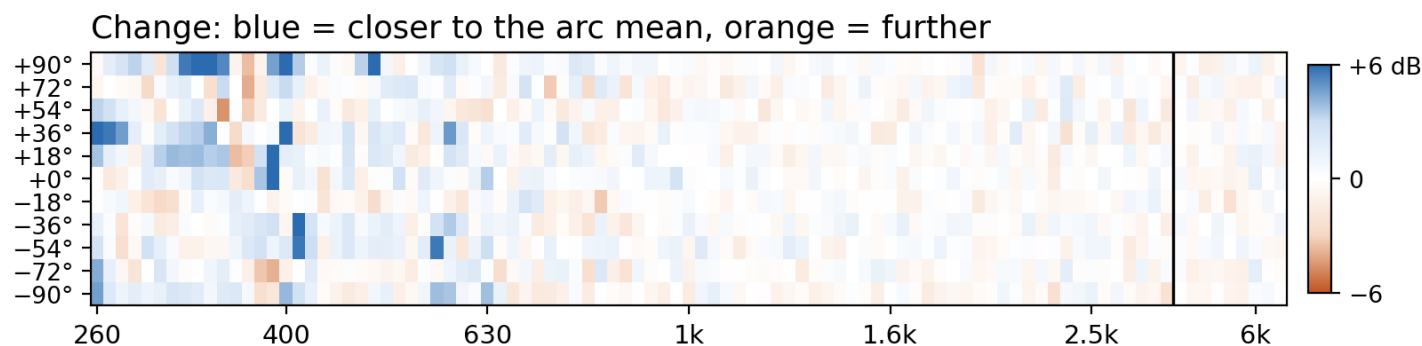
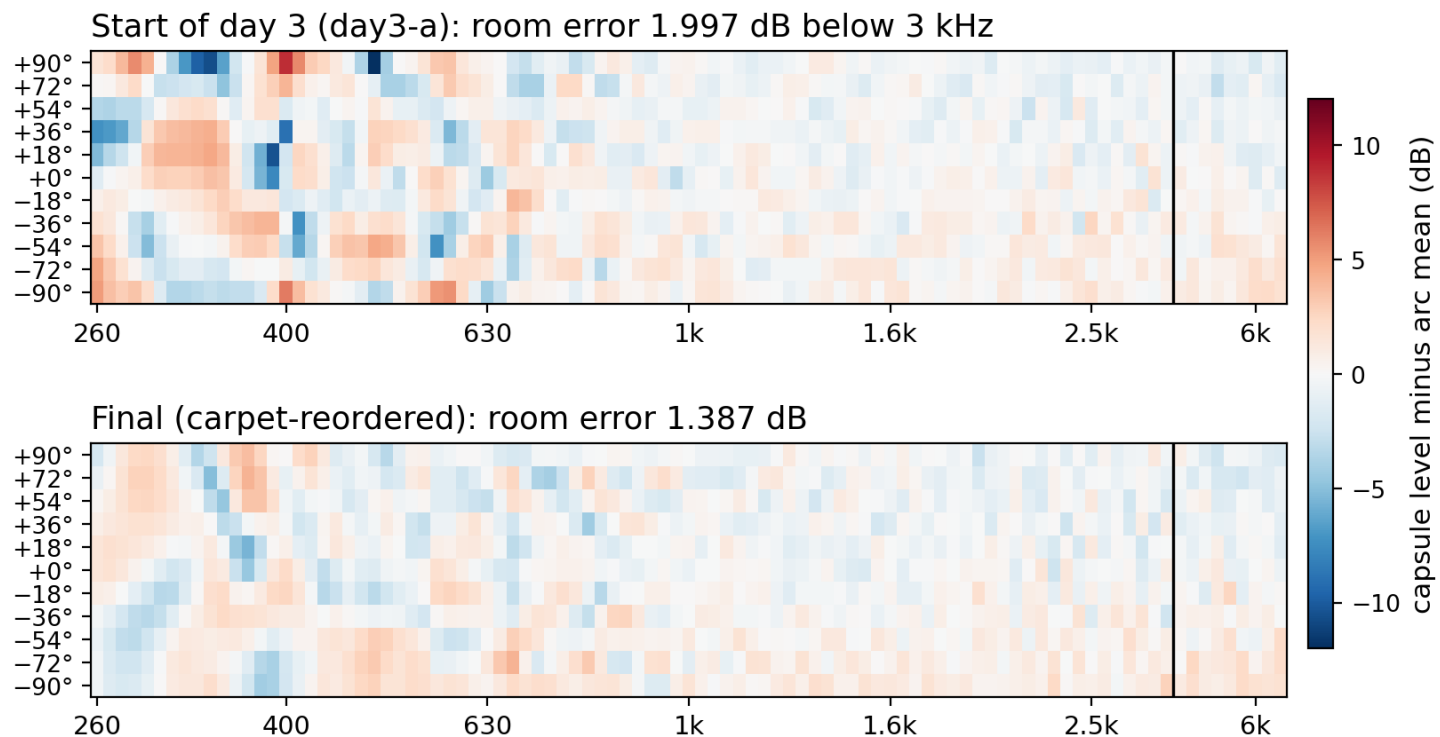
THE RESULT

The correction removes the microphone offsets

- **Top:** factory files. **Middle:** corrected. **Bottom:** what the correction changed.
- A stripe across all frequencies is one microphone reading too loud or too quiet. The middle row has none.
- The room pattern stays: whole rows move, not the frequency structure.

4.3 → 0.8 dB

gap between the loudest and quietest microphone for the same sound: factory files → corrected (start of day 3; final state 4.1 → 1.0 dB)



Frequency in Hz (k = kHz; black line: omitted 3-5 kHz). Top of each map = +90°.

THE CHAMBER

The treatments flatten the low end

Room error per band (dB, band in Hz)



■ Start of day 3 ■ Final

2.00 → 1.39 dB

room error below 3 kHz

We improved the low end (below 630 Hz) by 1.2 dB: the error fell from 3.0 to 1.8 dB, 40 % lower. Above 1 kHz only marginal changes.

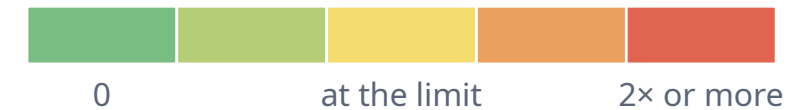
25 Sep 2026. The loudspeaker was moved during the day and returned to about its starting point; the position was not documented. Room error: rms difference of the eleven microphones from their average.

Band	Limit	Tones	Start of day 3		Final	
Hz	± dB	per band	worst mic, dB	cells in limit	worst mic, dB	cells in limit
250	1.5	3	6.81	36 %	1.88	52 %
315	1.5	8	3.46	28 %	0.89	56 %
400	1.5	8	3.46	41 %	1.41	59 %
500	1.5	8	1.93	41 %	1.99	59 %
630	1.5	8	0.93	43 %	1.49	61 %
800	1.0	8	0.87	60 %	1.12	55 %
1000	1.0	9	1.08	71 %	0.57	73 %
1250	1.0	8	0.54	83 %	1.02	73 %
1600	1.0	8	0.69	75 %	1.15	76 %
2000	1.0	8	0.66	84 %	0.77	82 %
2500	1.0	8	0.82	64 %	0.75	65 %
5000*	1.0	4	1.24	73 %	1.23	64 %
6300*	1.5	5	1.59	78 %	1.77	76 %

THE EVALUATION

Band by band against the ISO tolerance

Colour key

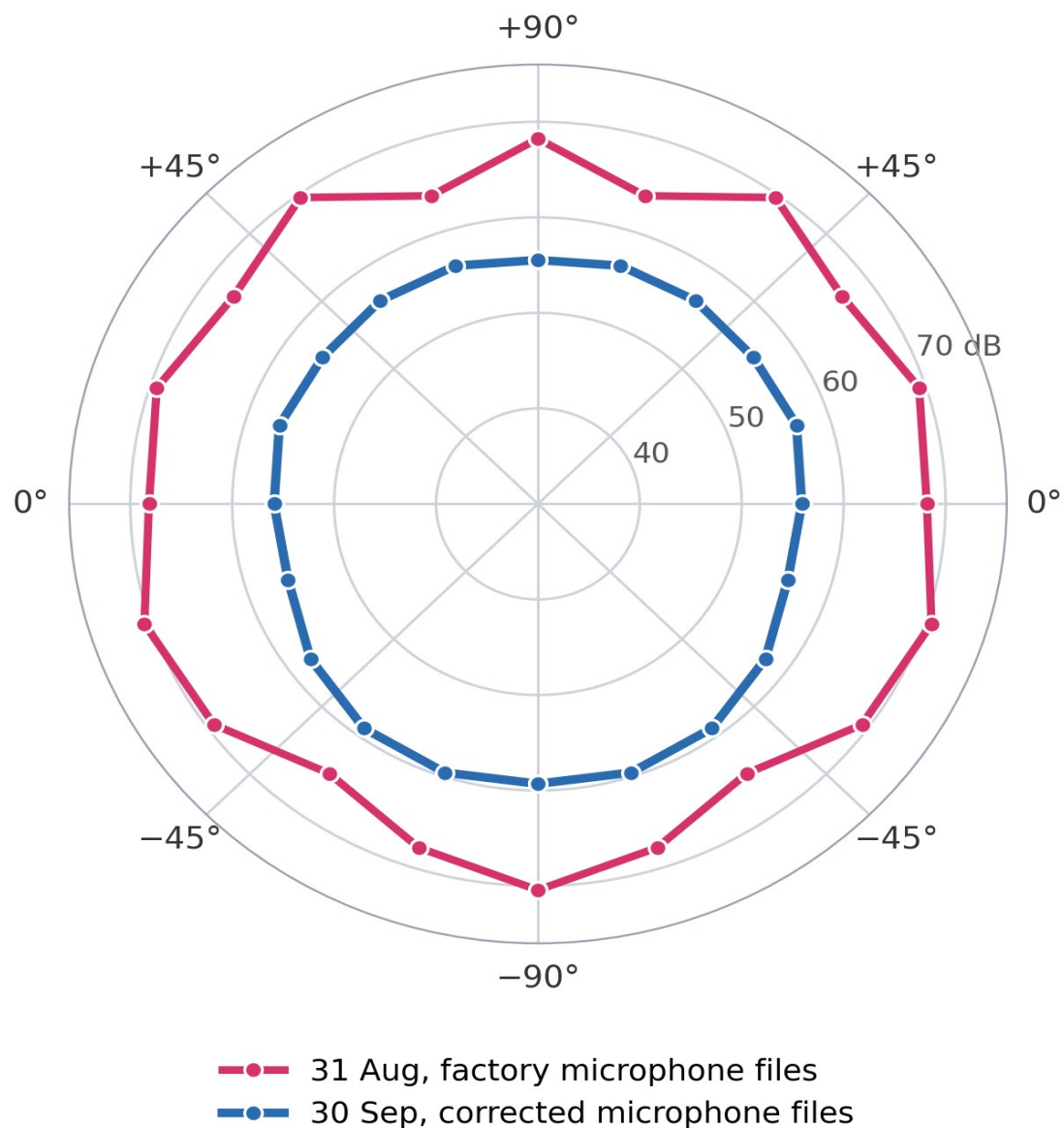


Worst microphone as a share of the limit. Cells: green 100 %, yellow 50 %, red 0 %.

Bands clearly over (of 13): **7 → 6**

Cells in limit (13 bands): **60 % → 66 %**

* Partial bands: only the tones above 5 kHz are measured (4 and 5 tones). 3–5 kHz is left out because of the loudspeaker: its cone has a rocking mode near 4.4 kHz. ISO values are an analogue, not a qualification of the room.



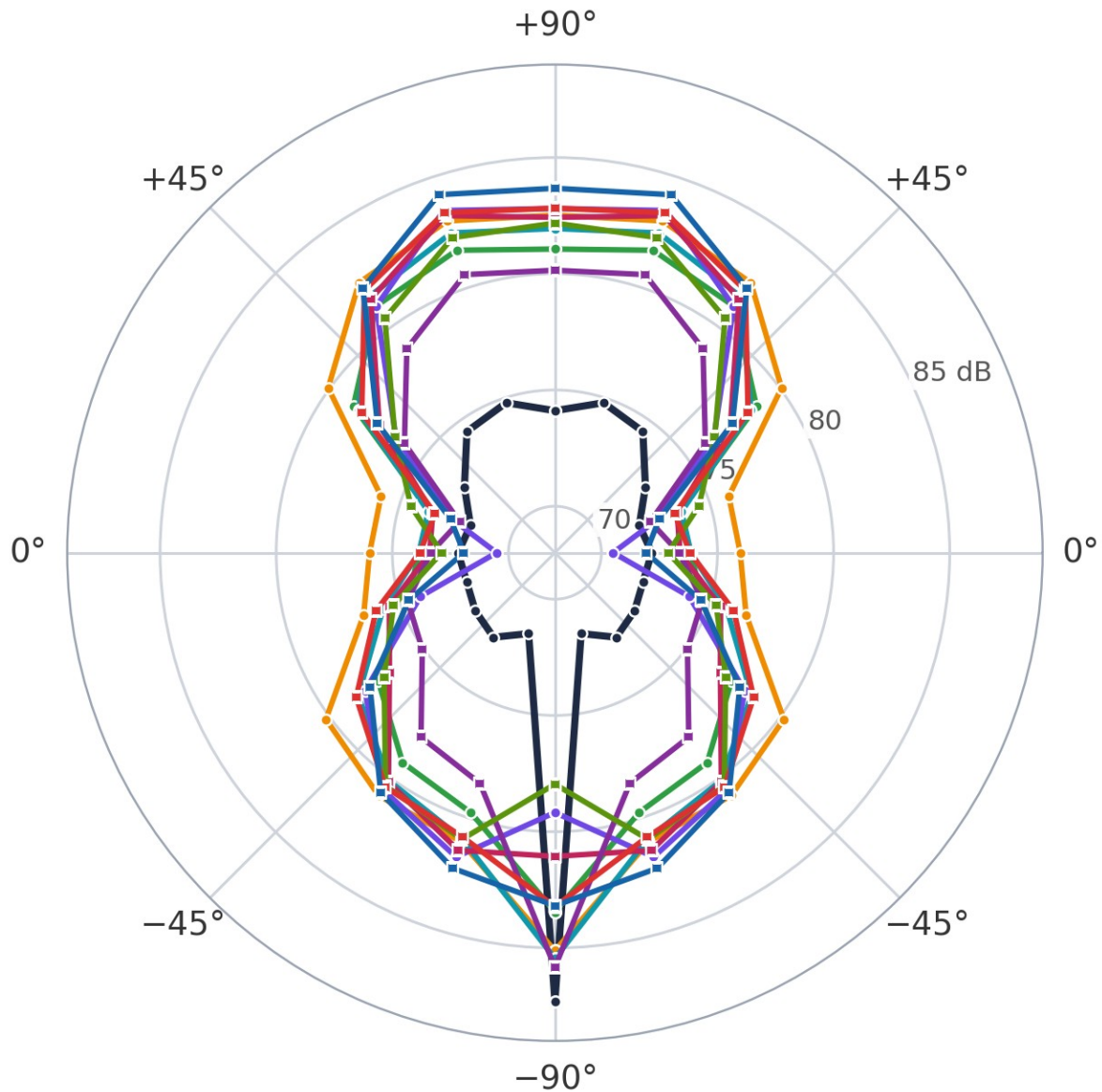
COMPARISON

31 Aug and 30 Sep on the same axes

- Red: 31 Aug, factory microphone files. Blue: 30 Sep, corrected files. Two different propellers.
- The propeller was mounted the other way round between the two, so the thrust points the other way. We do not expect this to change the sound.
- **The 30 Sep outline is smoother.**

6.7 → 4.0 dB

between the loudest and quietest microphone



THE MATERIALS

Materials at 2000 μs

Mean level. Round markers: 2 Oct. Square: 7 Oct.

● Bare	2 Oct	74.2 dB
● PU foam	7 Oct	77.6 dB
● Felt + rubber	2 Oct	78.2 dB
● PU foam, double layer	7 Oct	78.2 dB
● Cork	2 Oct	78.6 dB
● Inner felt + outer PU foam	7 Oct	78.8 dB
● Thinsulate, 8 layers	7 Oct	79.2 dB
● Felt	2 Oct	79.3 dB
● Thinsulate, 4 layers	7 Oct	79.4 dB
● Rubber	2 Oct	80.3 dB

+3.4 to +6.2 dB

louder than bare, on average (bare measured 2 Oct)